Bulk product upload - testing report

Testing report — Bulk product upload (Excel/CSV) with shop inventory update

Feature	Bulk product upload — design in docs/product-import-2026-10/README.md
Branch	`claude/amazing-ritchie-9vlq9f` — PR #173, merged into `staging` as `2886cf3`
Report date	15 Oct 2026; production smoke added 18 Oct 2026
Local environment	Node 22, PostgreSQL 16 (local), Next.js 16.3.6 dev server, Chromium (Playwright)
Test server (test.gokesari.com)	**Pass — 23 of 23 live cases** on 15 Oct 2026 (shop owner, admin, operator; desktop and the Android app's user agent). Migration `0077` confirmed on the test database (Test database workflow run 31: 80 migrations, newest `0077_product_bulk_import`). See §2.7.
Production (gokesari.com)	**Deployed 15 Oct 2026, 15:58 UTC** — release PR #179 (`acfaa6a`), migrations 0076–0078 applied by the build (`MIGRATE_ON_BUILD`). **Signed-out smoke checks pass (§2.8)**. **Signed-in smoke with a test shop: 4 of 4 pass on 18 Oct 2026 (§2.9)** — one product imported, stock and report checked, then deactivated by a Delete file.
Fixture	`tests/fixtures/bulk_upload/PUZZLES.xlsm` — **a reconstruction** of the Amazon PUZZLES template to the layout the brief verified (309 columns, rows 1–7 as described, data from row 8, VBA project included). The original file was not available to this session.

Results columns: **Local** = run in this session (automated suite or browser); **Test** / **Prod** = run on test.gokesari.com / gokesari.com. "Not run" means exactly that — nothing is claimed for an environment it was not run on.

1. Summary

Suite	Cases	Local	Test	Prod
Parser and normalisation (unit)	18	**18 pass**	Not run	Not run
Import service and API (integration, real PostgreSQL)	23	**23 pass**	Not run	Not run
Load: 5,000 rows	1	**pass** (parse + preview 1.8 s; import 94.7 s)	Not run	Not run
Browser walk-through (desktop, phone, admin, operator)	10	**10 pass**	**10 pass** (by the live run, §2.7)	Not run
Live functional run on the site (`scripts/test-site/product-import-check.mjs`)	23	—	**23 pass**	Smoke subset only (§2.9)
Production smoke, signed out (§2.8)	13	—	—	**13 pass**
Production smoke, signed in with a test shop (§2.9)	4	—	—	**4 pass**
Mobile app (unit, typecheck, lint)	26 tests	**pass**	—	—
Full regression suite (all existing unit + integration tests)	1,811	**all pass** after D-9, re-checked after merging `staging` (see §4)	Not run	Not run

Suite	Cases	Local	Test	Prod
Environment end-to-end (web, Android, iOS on real accounts)	9	—	**4 pass** (E-01–E-03, E-06); E-04/E-05 need a phone; E-09 see §2.6	**E-06, E-07, E-08 pass** (E-07 and E-08 signed in, §2.9)

2. Test cases

2.1 Parser — `tests/unit/product-import-parser.test.ts`

ID	Case	Steps	Expected	Actual (local)	Local	Test	Prod
P-01	PUZZLES.xlsm layout	Parse the fixture	Sheet `Template`, 309 columns, mapped by attribute key; reading stops at Template	As expected; sheets read: Instructions, Data Definitions, Template	Pass	Not run	Not run
P-02	Example and notice rows	Same	Rows 6 (ABC123) and 7 (notice) left out; first data row 8	`skippedTemplateRows = [6, 7]`; first row 8	Pass	Not run	Not run
P-03	Row 8 values	Same	Product type PUZZLES, brand Generic, HSN 95032000 (External Product Information, Entity “HSN Code”), Quantity 25, MRP ₹999, price ₹649	As expected	Pass	Not run	Not run
P-04	Repeated columns → arrays	Same	Bullet Point ×5, Other Image URL, Material, Theme, Safety Warning as arrays	As expected	Pass	Not run	Not run
P-05	Lossless	Same	Every non-empty cell in raw (by key) and in attributes by section; B2B kept apart	As expected	Pass	Not run	Not run
P-06	Indian numbers	`1,29,999.50`, `₹1,299.00`, `Rs. 99/-`, `INR 1299`, `99-120`, `-5`	Numbers / paise; ranges and negatives invalid	As expected	Pass	Not run	Not run
P-07	Dates	serial 46300 / 45658, `15/10/2026`, `15-10-26`, ISO, `15 Oct 2026`, `31/02/2026`	ISO dates; impossible date invalid	As expected	Pass	Not run	Not run
P-08	Units	mm, Centimeters, `12 cm`, inches, Kilograms, Grams, stone	cm / g; unknown unit invalid	As expected	Pass	Not run	Not run
P-09	Booleans	Yes/y/TRUE/1, No/n/false/0, maybe	true / false / invalid	As expected	Pass	Not run	Not run

ID	Case	Steps	Expected	Actual (local)	Local	Test	Prod
P-10	Other product types	GROCERY-style keys under misleading labels	Mapped by key	As expected	Pass	Not run	Not run
P-11	CSV	Semicolon CSV with BOM and `₹45`	Read; format CSV	As expected	Pass	Not run	Not run
P-12	GoKesari template	Build template, read it back, parse it	All columns, Instructions (“ALL COLUMNS ARE OPTIONAL”), dropdowns (Barcode Type, Unit, Action), example row never imported	As expected	Pass	Not run	Not run
P-13	Report formula injection	Cell `=HYPERLINK(…)`	Written as `'=HYPERLINK(…)`	As expected	Pass	Not run	Not run
P-14	Not a spreadsheet	`.xls` (OLE), `.pdf`	422 with a readable message	As expected	Pass	Not run	Not run
P-15	Planned stock	Set / Add / blank / new / deactivate	4 / 17 / unchanged / 0 / 0	As expected	Pass	Not run	Not run
P-16	All-optional rows (parser)	Only a name, only a barcode, an empty row, only SKU + Quantity	3 rows read, 1 empty counted	As expected	Pass	Not run	Not run
P-17	SSRF: URL checks	127.0.0.1, 10.x, 169.254.169.254, 192.168.x, [::1], localhost, file:, ftp:, user:pass@	All refused	As expected	Pass	Not run	Not run
P-18	SSRF: DNS to a private address	Host name resolving to 10.1.2.3	Refused at connect time (BLOCKED)	As expected	Pass	Not run	Not run

2.2 Import service and API — `tests/integration/product-import.test.ts`

ID	Case	Steps	Expected	Actual (local)	Local	Test	Prod
I-01	PUZZLES.xlsm end to end	Owner uploads, previews, imports (Set)	5 rows → 5 created; Pending master products; HSN, GTIN on product; MRP on the listing only (master MRP null); prices live; stock 25 / 1200 / 0 / 0 / 7; “ten” → warning, stock 0; ledger rows with source bulk_upload, job and user; owner notified in app	As expected	Pass	Not run	Not run

ID	Case	Steps	Expected	Actual (local)	Local	Test	Prod
I-02	Result report	Download CSV and XLSX	Original columns + `status` + `messages`	As expected	Pass	Not run	Not run
I-03	Match by SKU	Import, then a file with the same SKU (other case)	Updated; blank name cell does not wipe the name	As expected	Pass	Not run	Not run
I-04	Match by barcode	Master product with GTIN; file with that barcode	Linked; no new master row; shop carries the category	As expected	Pass	Not run	Not run
I-05	Match by name + brand	Product made by shop A; shop B's file with the same name/brand (different case/spacing)	"Matched by name" warning; one master row	As expected	Pass	Not run	Not run
I-06	New → Pending, duplicate check	No match	Pending product; manual "Create New Product" with that name reuses it	As expected	Pass	Not run	Not run
I-07	All optional	Rows: only name / only barcode / empty / only SKU + Quantity	3 rows, 1 skipped; generated SKU; "not orderable until stock is set"; stock 42 and nothing else changed for the SKU row	As expected	Pass	Not run	Not run
I-08	Set mode	Stock 10, file 3	3	3	Pass	Not run	Not run
I-09	Add mode	Stock 10, file 5	15	15	Pass	Not run	Not run
I-10	Blank / negative quantity	Blank; then −4	Unchanged both times; warning on −4	As expected	Pass	Not run	Not run
I-11	Duplicate SKUs in one file	Rows (12, 4) and (blank, 6)	Set → 6, price 12 kept; Add → +10; earlier row "merged"	As expected	Pass	Not run	Not run
I-12	Concurrent orders during an import	Stock 20; Add 5 while 6 orders consume 1 each	19; ledger sums to the balance	As expected	Pass	Not run	Not run
I-13	Listing actions	Upsert; Edit (no match / match); Delete; Upsert again	Edit-no-match skipped; Delete → inactive, stock 0, not deleted, master kept; Upsert reactivates	As expected	Pass	Not run	Not run
I-14	Operator prices	Operator imports a new price	Price unchanged; 2 price requests (online, in-shop) pending; stock updated; operator and owner notified	As expected	Pass	Not run	Not run

ID	Case	Steps	Expected	Actual (local)	Local	Test	Prod
I-15	Price above MRP	MRP 100, price 120	Price not applied, warning	As expected	Pass	Not run	Not run
I-16	Images	Two rows share a URL; one SSRF URL (169.254.169.254)	URL downloaded once; 2 photos saved, 1 refused as a row warning; products created	As expected	Pass	Not run	Not run
I-17	Other shop	Owner A uploads for shop B; cancels B's job	403; 404	As expected	Pass	Not run	Not run
I-18	Oversized / wrong type	100 kB limit; `.pdf` ; fake `.xlsx`	422; 422; preview fails with a readable message	As expected	Pass	Not run	Not run
I-19	Macros never run	`.xlsm` with a VBA project (Auto_Open + Shell)	Imported as data; "macros… not run" note	As expected	Pass	Not run	Not run
I-20	History, duplicate file, cancel, retention	Same file twice; cancel; age 91 days; purge	"Already imported" with the first job; cancelled; file and rows purged, summary kept	As expected	Pass	Not run	Not run
I-21	API v1 end to end	Upload (multipart) → poll → rows (warnings) → commit → report → history → template	202 / preview_ready / rows / 202 / completed / CSV attachment / 1 import / template file	As expected	Pass	Not run	Not run
I-22	API access	Signed out; other owner	401; 403	As expected	Pass	Not run	Not run
I-23	Regression: existing flows	Manual create product; add category to shop	Unchanged behaviour (Pending product, price; category fills inventory)	As expected	Pass	Not run	Not run

2.3 Load — `tests/integration/product-import-load.test.ts`

ID	Case	Expected	Actual (local)	Local	Test	Prod
L-01	5,000 new products (166 KB file), Set mode	All created, ledger rows for non-zero quantities, no UI timeout (202 + polling)	Upload 33 ms; parse + preview 1.8 s; import 94.7 s (≈19 ms/row); 5,000 created, 0 failed	Pass	Not run	Not run

First run took 158 s for the import; new listings now get their price, stock and order limit in the insert itself (same checks, same ledger row), which brought it to 95 s. On the hosted database each query costs more, so a full file of new products will take several minutes there — it runs after the response, with a progress bar.

2.4 Browser walk-through (local dev server, seeded demo data)

Script: Playwright (Chromium). Screenshots in docs/product-import-2026-10/screenshots.

ID	Role / device	Steps	Expected	Actual	Local	Test	Prod
U-01	Shop owner, desktop	Inventory page	“Bulk upload (Excel/CSV)” in the Inventory menu strip and Seller Central sidebar	Shown (01)	Pass	Pass (PI-01)	Not run
U-02	Shop owner, desktop	Open Bulk upload	Drop zone, template download, limits, help	Shown (02)	Pass	Pass (PI-02)	Not run
U-03	Shop owner, desktop	Upload PUZZLES.xlsm	Checking… then preview: totals, stock mode, table with Qty current → new and warnings	Shown (03)	Pass	Pass (PI-03)	Not run
U-04	Shop owner, desktop	“Only rows with warnings”	Filtered table	Shown (04)	Pass	Pass (PI-04)	Not run
U-05	Shop owner, desktop	Import	Progress, then result totals, row outcomes, image status, history	Shown (05)	Pass	Pass (PI-05)	Not run
U-06	Shop owner, desktop	Download report	`PUZZLES_result.xlsx`	Downloaded (sample)	Pass	Pass (PI-06)	Not run
U-07	Shop owner, desktop	Upload the same file again	“This file was already imported on …”	Shown (06)	Pass	Pass (PI-07)	Not run
U-08	Shop owner, phone 390 px with the app’s user agent	Upload, preview (cards), Add mode, import	Card layout; import completes	Shown (07–09); second shop matched the first shop’s products by barcode and by name + brand	Pass	Pass (PI-12 cards; Add mode PI-10)	Not run
U-09	Admin	Product master hub → Bulk upload → pick shop → import	Tile, shop picker, import for that shop	Shown (10–12)	Pass	Pass (PS-01–04, admin)	Not run
U-10	Operator	Catalogue hub → Bulk upload → pick shop → import	Tile, picker, import; prices proposed to the owner	Shown (13–14)	Pass	Pass (PS-01–04, operator)	Not run

2.5 Mobile apps

ID	Case	Expected	Actual	Local	Test	Prod
MOB-01	Navigation policy	Bulk-upload pages, API and report download stay in the app	`mobile/src/__tests__/navigation-policy.test.ts` — pass	Pass	—	—
MOB-02	App checks	Typecheck, lint, 26 unit tests	Pass	Pass	—	—
MOB-03	Android build (APK)	Version 1.1.0, versionCode 2; preview + production APKs	Built by the PR’s “Android APK” workflow — see §5	See §5	—	—
MOB-04	Android release AAB	Play Store build	**Not run** — needs EAS (Expo account / token)	Not run	—	—
MOB-05	iOS archive / TestFlight	Version 1.1.0, buildNumber 2	**Not run** — needs EAS and an Apple Developer account	Not run	—	—

ID	Case	Expected	Actual	Local	Test	Prod
MOB-06	Document picker on a phone	Android system picker (Files, Drive, WhatsApp, e-mail) / iPhone Files offers .xlsx/.xlsm/.csv	The upload is a web file input with explicit MIME types (including the macro-enabled type and `application/octet-stream` for files saved from WhatsApp); react-native-webview hands it to the system picker. **Needs a device on the test site**	Not run	Not run	Not run
MOB-07	Push notification on completion	Arrives on the phone	In-app notification written and a PUSH copy queued; no push provider is configured on the platform yet	Not run	Not run	Not run

2.6 Environment end to end (the brief's §8, steps 1–2)

ID	Where	Steps	Test	Prod
E-01	Web, shop account	Upload PUZZLES.xlsm + the GoKesari template, preview, import (Set), check stock = Quantity, download report	**Pass** (PI-01–PI-15)	—
E-02	Web, admin account	Pick a shop, import, check the owner is notified	**Pass** (PS-01–04; owner's in-app notifications list the admin's imports)	—
E-03	Web, operator account	Pick a shop, import, check prices go to the owner	**Pass** (PS-03: "1 price change(s) were sent to the shop owner for approval"; the admin's price applied at once)	—
E-04	Android app (preview build)	Pick the file from Files / Drive / WhatsApp, preview cards, import, share the report	Partly: the app's user agent at phone width shows the card preview (PI-12). The system file picker needs a real phone — **not run**	—
E-05	iPhone	Same, from Files	Not run	—
E-06	Migration	`0077` applied by the Test database workflow	**Pass** (run 31: 80 recorded, newest `0077_product_bulk_import`; backup taken first)	**Pass** — applied by the gokesari.com build (`MIGRATE_ON_BUILD`; the build stops before deploying if a migration fails, and the new routes answer)
E-07	Prod smoke	One small import into a designated test shop; check stock and the report	—	**Pass** (18 Oct 2026, §2.9): one-row import into the test shop, stock 3 = file Quantity, report row "created", then the Delete file deactivated it
E-08	Prod clean-up	Delete-action file deactivates the test products	—	**Pass** (18 Oct 2026, PROD-04 in §2.9): the smoke product is hidden with stock 0
E-09	Regression on the site	Manual add product, add category, Excel price upload	Not run on the site (covered locally by I-23 and the full suite, §4)	Not run

2.7 Live run on test.gokesari.com (15 Oct 2026)

Script: `scripts/test-site/product-import-check.mjs` (Chromium, signed in with `scripts/test-site/signin.mjs` and the e-mailed code). Shop: **E2E Referral MV2SJM8P-3** (`e1dc0a8d-…`, the shop owner test account). Accounts: `+gk-shop` (owner), `+gk-admin`, `+gk-ops`. Screenshots and the raw JSON results are in docs/product-import-2026-10/test-site-2026-10-15/.

ID	Case	Result	Detail
PI-01	Inventory menu has the Bulk upload (Excel/CSV) tile	**Pass**	Link to `/shop/product-import` (T01)
PI-02	Bulk upload page and the GoKesari template download	**Pass**	`GoKesari_Product_Upload_Template.xlsx` (T02)
PI-03	Upload PUZZLES.xlsm in the browser: preview, 309 columns, rows 6–7 left out, macros not run	**Pass**	(T03)
PI-04	Only rows with warnings	**Pass**	1 row (“ten” is not a whole number) (T04)
PI-05	Import (Set) completes with no failed row	**Pass**	First run: 5 created. Re-import: 5 updated, 0 failed (T05)
PI-06	Result report download (Excel)	**Pass**	`PUZZLES_result.xlsx`
PI-07	Same file again: “already imported” prompt	**Pass**	(T06)
PI-08	Import history (API)	**Pass**	All the shop’s imports listed
PI-09	GoKesari template file: new products, stock = file Quantity	**Pass**	3 created, 0 failed
PI-10	Stock: Set → 11; Add 4 → 15; preview shows current → new	**Pass**	11 → 15
PI-11	Blank and negative quantities leave stock unchanged; Edit with no match skipped	**Pass**	Stock still 15
PI-12	Phone width with the Android app’s user agent: preview as cards	**Pass**	(T07)
PI-13	Delete action soft-deactivates (stock 0, listing kept)	**Pass**	2 deactivated (the run’s own test products)
PI-14	Owner notified in app when an import finishes	**Pass**	“Bulk product upload finished” per import
PI-15	Another shop’s id is refused	**Pass**	404
PS-01 (admin)	Bulk upload tile on the staff board	**Pass**	Product master hub (admin-S01)
PS-02 (admin)	Shop picker	**Pass**	(admin-S02)
PS-03 (admin)	Import for the chosen shop	**Pass**	Created; price applied at once (admin-S03)
PS-04 (admin)	Clean-up: Delete file deactivates the product	**Pass**	
PS-01 (operator)	Bulk upload tile on the staff board	**Pass**	Catalogue hub (ops-S01)
PS-02 (operator)	Shop picker	**Pass**	(ops-S02)
PS-03 (operator)	Import for the chosen shop	**Pass**	Created; price sent to the owner for approval (ops-S03)
PS-04 (operator)	Clean-up: Delete file deactivates the product	**Pass**	

Notes from the live run: - Image URLs in the PUZZLES fixture point at `images.example-cdn.com`, which answers with an HTML page. Each image was refused with a row warning (“not a JPEG, PNG or WebP image (text/html)”) and the products were still created, as designed. A file with real image URLs was not tried on the site. - After “Completed” appears, the result rows replace the preview rows on the next refresh, about 3 s later on the test site. Not a defect; the script waits for it. - Earlier attempts of the same run failed only because of the test script: the shop id was read before the history list had refreshed, and the “Import it again?” browser confirmation was dismissed instead of accepted. Both were fixed in the script. The final run above passed in full. A preview left open by an interrupted attempt was cancelled. - Test data left on the test shop (test site only): the five PUZZLES products, and per run “Live Test Product … C” (generated `GK-…` SKU, stock 4) — both still listed. Each run’s `LIVE-…-A` / `LIVE-…-B` and every staff `STAFF-…` product were deactivated by the run’s own Delete files.

2.8 Production smoke, gokesari.com (15 Oct 2026, after release #179)

Signed out, no writes. A route that answers 401 or redirects to sign-in, instead of 404, shows the new code is live. The deploy only builds after its migrations succeed, so these answers also show 0076–0078 are applied.

ID	Check	Result
PP-01	`/shop/product-import` (owner page)	**Pass** — 307 to /signin
PP-02	`/admin/product-import` (staff page)	**Pass** — 307 to /signin
PP-03	GET `/api/v1/product-imports/template`	**Pass** — 401 (was 404 before the release)
PP-04	GET `/api/v1/product-imports/shops`	**Pass** — 401
PP-05	GET `/api/v1/shops/{id}/product-imports` (history)	**Pass** — 401
PP-06	GET `/api/v1/product-imports/{id}` (status)	**Pass** — 401 (one request timed out at the network; three retries 401)
PP-07	GET `…/{id}/rows`	**Pass** — 401
PP-08	GET `…/{id}/report`	**Pass** — 401
PP-09	POST `/api/v1/shops/{id}/product-imports` (upload)	**Pass** — 401
PP-10	POST `…/{id}/commit`	**Pass** — 401
PP-11	POST `…/{id}/cancel`	**Pass** — 401
PP-12	Existing pages `/`, `/shops`, `/signin`	**Pass** — 200; `/shop` 307 to sign-in
PP-13	Inventory & Accounting smoke (`scripts/test-site/prod-smoke.mjs`, released with it)	**Pass** — 11/11 (module pages, API 401s, crons 403, share target, manifest, existing pages, desktop and Android UI with no page errors)

The signed-in upload and import were run later with a test shop (§2.9).

2.9 Production smoke, signed in, with a test shop (18 Oct 2026)

Test shop. A shop was registered on gokesari.com for this run: “GoKesari QA Test Shop (internal testing - do not order)”, id `3a0557f6-42bc-4ee4-81e9-c91109ece9f1`. It was registered with `POST /api/shops` by the test account `sanjaymorankar+gk-shop@gmail.com`, which became its owner. No registration fee was paid, so the shop stays **Pending approval**. It is not shown in public search (checked after the run) and customers cannot order from it. **It must never be approved.**

Run. `node scripts/test-site/product-import-check.mjs https://gokesari.com <state> smoke <dir> 3a0557f6-… --prod-smoke`, signed in as the shop owner, 16:01–16:03 UTC. Evidence (screenshot, results JSON, the import’s result CSV) is in prod-smoke-2026-10-18/.

ID	Check	Result
PROD-01	The owner’s bulk upload page `/shop/product-import` loads	**Pass**
PROD-02	One-row file `smoke-MVE0J1I1.xlsx` (SKU `SMOKE-MVE0J1I1`, price 10, Quantity 3) is uploaded and imported; listing stock = file Quantity	**Pass** — job `485e6835-5ca4-4e7e-b7b0-6d855e55e39c`, 1 created, 0 failed, stock 3
PROD-03	Result report has the row	**Pass** — status `created`; message “No category given; the product goes in”General” (uncategorised)”
PROD-04	Clean-up: a Delete file deactivates the smoke product	**Pass** — job `smoke-del-MVE0J1I1.xlsx 1` deactivated, 0 failed; the listing is hidden with stock 0

Checked afterwards: the owner received both “Bulk product upload finished” notifications. The import history shows the import, the Delete file and a cancelled preview (`smoke-check-…`). That preview is part of PROD-02: it re-reads the SKU to confirm the stock now held, then is cancelled, so nothing is written. The shop is still Pending approval.

Left on production by this run (harmless, recorded so nobody is surprised):

- the pending test shop above and its deactivated listing `SMOKE-MVE0J1I1`;
- the master product “Smoke Test MVE0J1I1 (delete)”, Pending approval in the catalogue (an upload never approves a new master product). Reject or delete it in the admin catalogue review;
- the test account `sanjaymorankar+gk-shop@gmail.com`, now a shop owner on production.

Script fix found by this run. In smoke mode PROD-01 always opened the staff page `/admin/product-import`, which a shop owner cannot use. It now opens `/shop/product-import` for a shop owner and keeps the staff page for staff.

3. Defects found and fixed during this work

#	Found by	Defect	Fix
D-1	I-01 and others	Finishing an import failed with a SQL syntax error when there were no job messages (empty array interpolated as `()`)	Messages appended in code, not SQL
D-2	I-01	Rows with a brand not in the brand list failed (`""` used as a brand id)	Missing brand cached as “none”
D-3	I-03	Re-importing a SKU in another letter case overwrote the stored SKU	A listing’s SKU is set once; same SKU in another case is the same SKU
D-4	P-11	A CSV with the template’s headings was reported as the GoKesari template	Format detection checks the file type first
D-5	Lint	The screen loaded data synchronously inside effects (React lint error)	Loads moved to timer callbacks
D-6	U-05	Image failures showed raw network errors (`getaddrinfo ENOTFOUND …`)	Plain reasons (“the web address could not be found”)
D-7	U-05	After an import the upload box re-appeared above the result	Shown again only on “Upload another file”
D-8	L-01	5,000 rows took 158 s	New listings written in one insert (95 s)

4. Regression suite

Full `npm test` (all unit and integration tests, local PostgreSQL), after the feature: **1,811 tests pass, 0 fail** (unit + integration, local PostgreSQL 16), re-run after merging `staging` (#168, #169, #171) and renumbering the migration to `0077`. CI on PR #173 (test, mobile, Android APK) was green before the merge.

5. App build numbers

App	Version	Build number	Where
Android	1.1.0	versionCode 2	APK (preview + production): “Android APK” workflow run 19 on PR #173 — artifacts on the run
Android (Play Store AAB)	1.1.0	(EAS auto-increments)	Not built — needs EAS
iOS	1.1.0	buildNumber 2	Not built — needs EAS + Apple Developer account

6. Sign-off

Environment	Result	By	Date
Local	Pass (automated + browser)	Claude Code	15 Oct 2026
test.gokesari.com	Pass — 23/23 live cases (§2.7)	Claude Code	15 Oct 2026
gokesari.com	Deployed; signed-out smoke 13/13 pass (§2.8)	Claude Code	15 Oct 2026
gokesari.com	Signed-in smoke with a test shop 4/4 pass (§2.9)	Claude Code	18 Oct 2026